Douglas McDonough

Selected Research Experience

The following document highlights my successful design and prototyping experience as an undergraduate research assistant in the Advanced Robot Manipulator Lab at Stevens Institute of Technology (May 2024 - Present).

Project Scope: To construct a novel instrument module with a tendon-driven continuum probe for minimally invasive surgical applications, and evaluate efficacy with the integration of a 7-DOF robot arm controlled with ROS2.

Design Work: The pictures below depict selected SolidWorks part models, physical prototypes, and various diagrams that I designed, constructed, and developed.

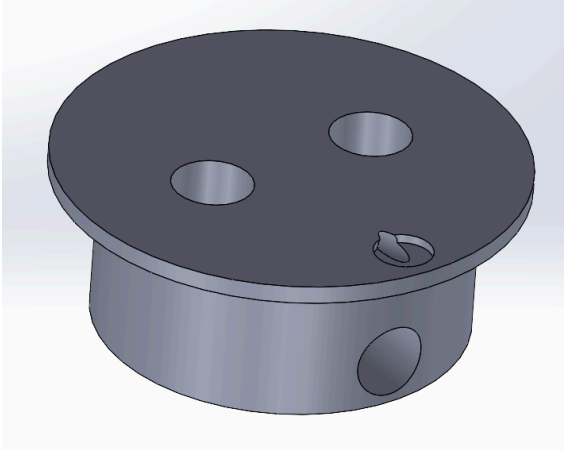


Figure 1. Pulley SolidWorks model

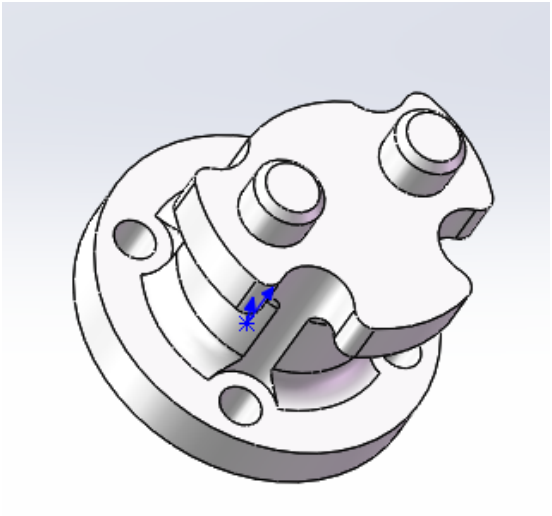


Figure 2. Servo-Pulley Connector SolidWorks model

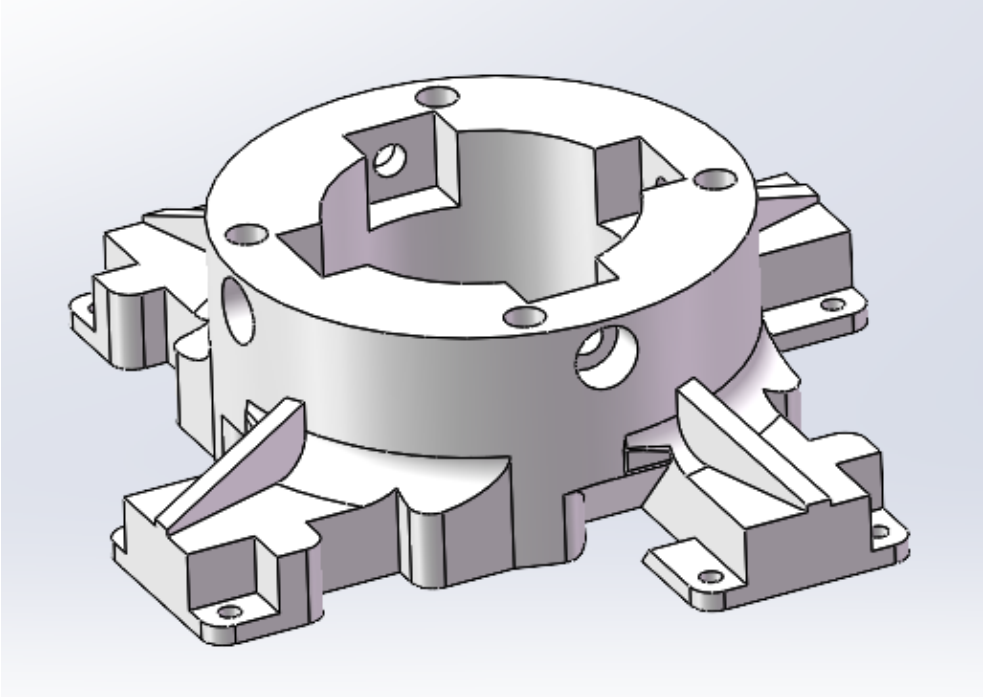


Figure 3. Servo Lower Adapter SolidWorks model

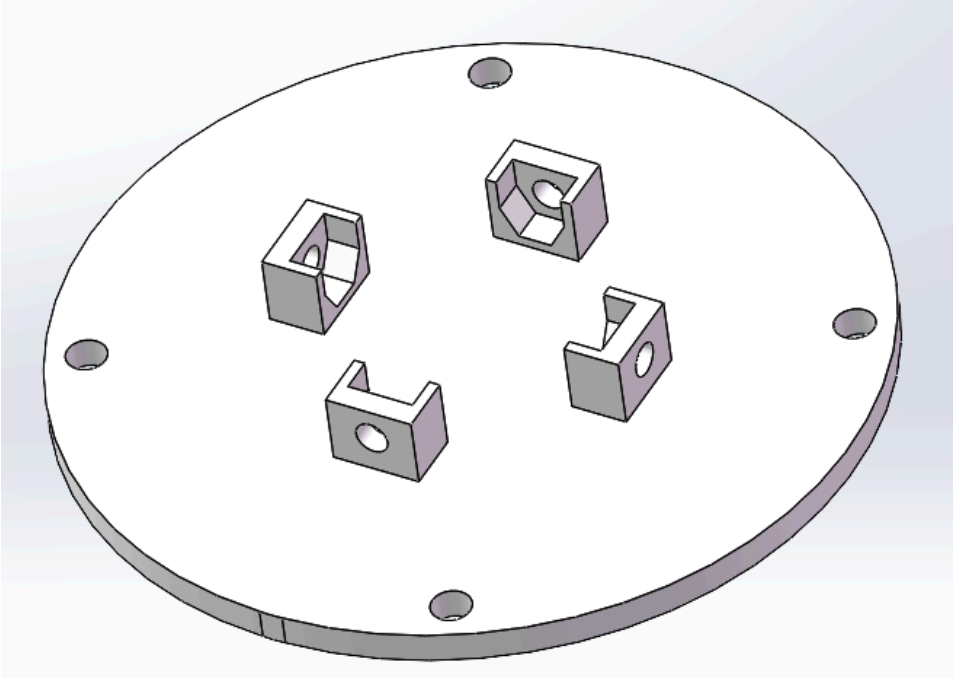


Figure 4. Servo Upper Adapter SolidWorks model

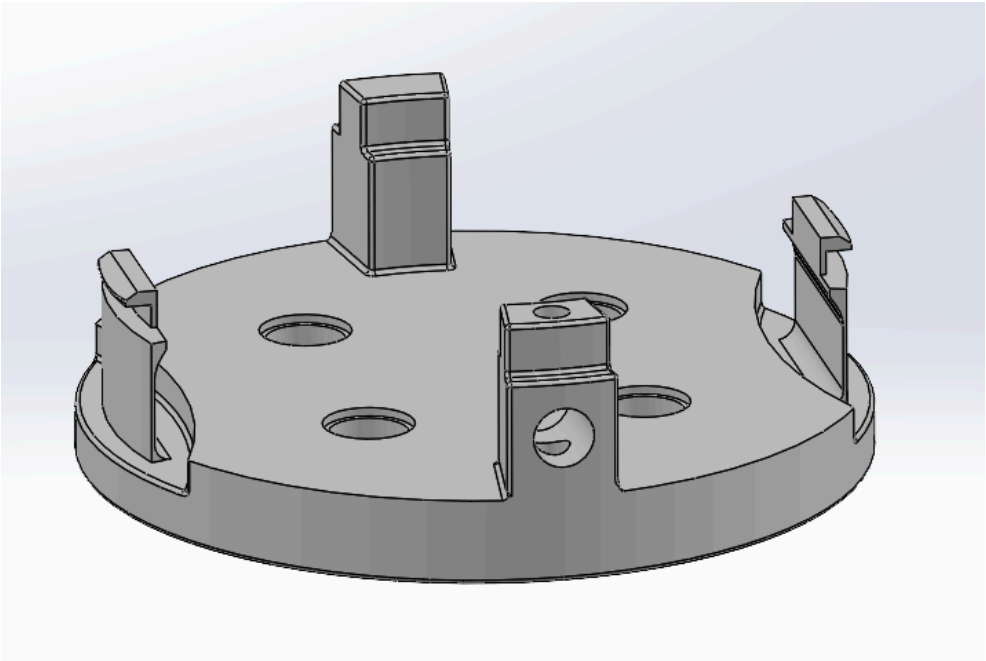


Figure 5. Instrument Module Bottom Plate SolidWorks model

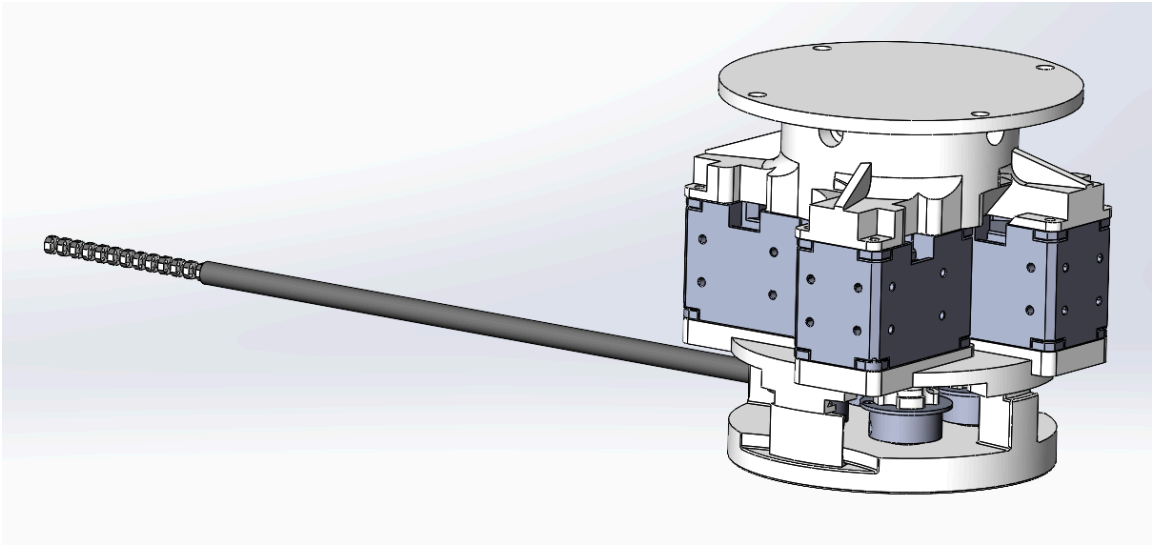


Figure 6. Final Instrument Module design with continuum probe Solidworks model



Figure 7. Partially constructed Instrument Module

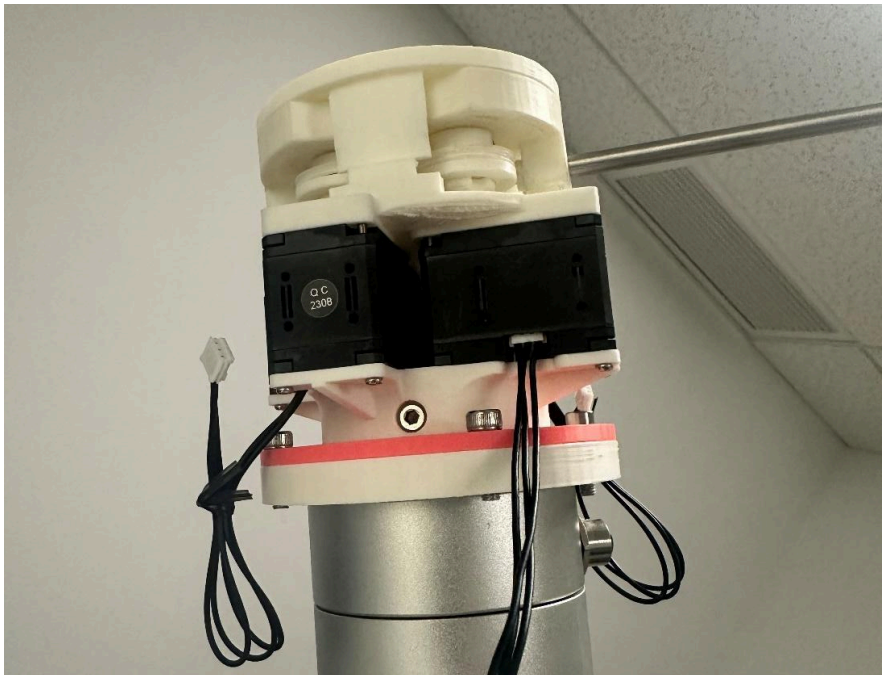


Figure 8. Constructed Instrument Module on 7-DOF robot arm

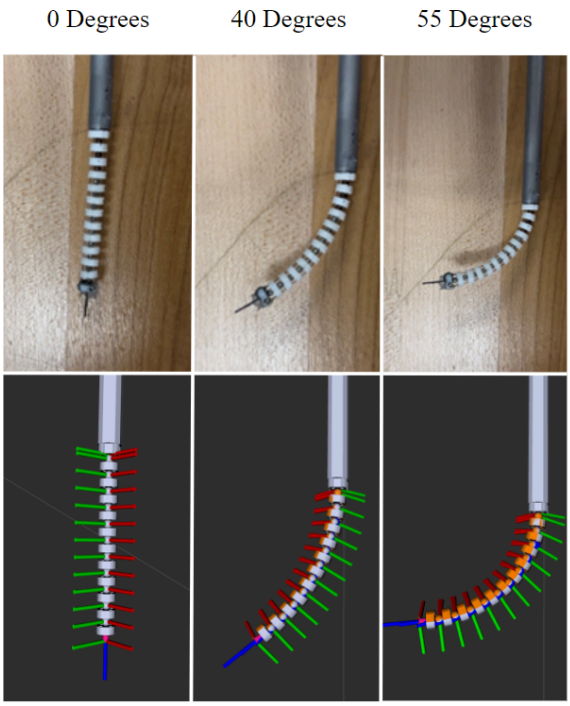


Figure 9. Diagram comparing the continuum probe tip angle between prototype and simulation space (ROS2)

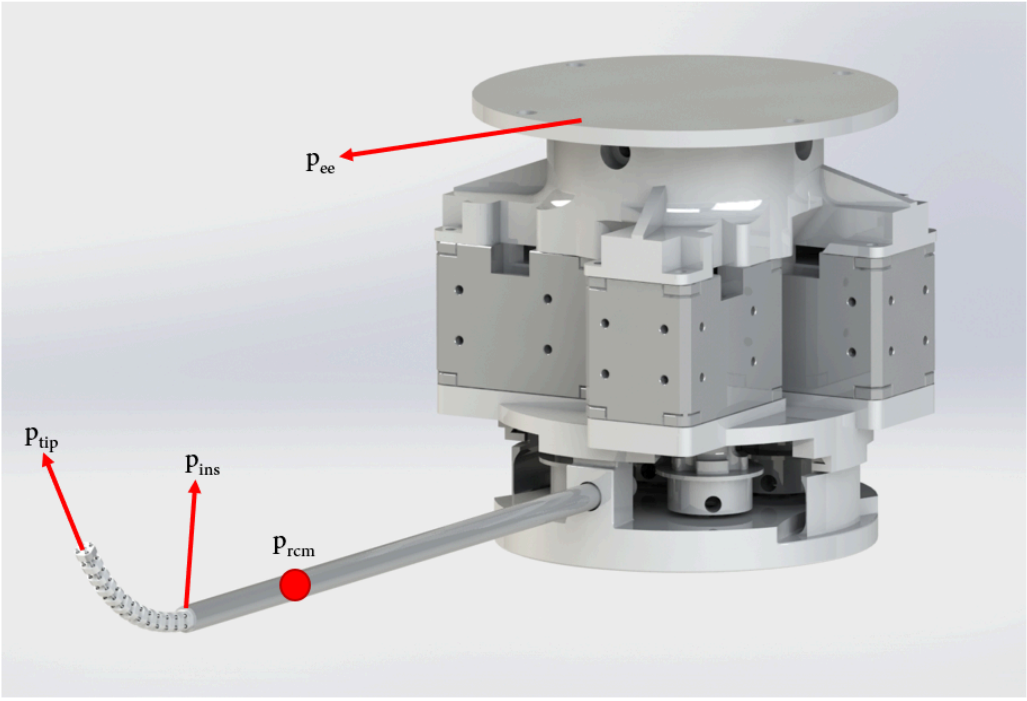


Figure 10. Diagram depicting final rendering with relevant vector markers

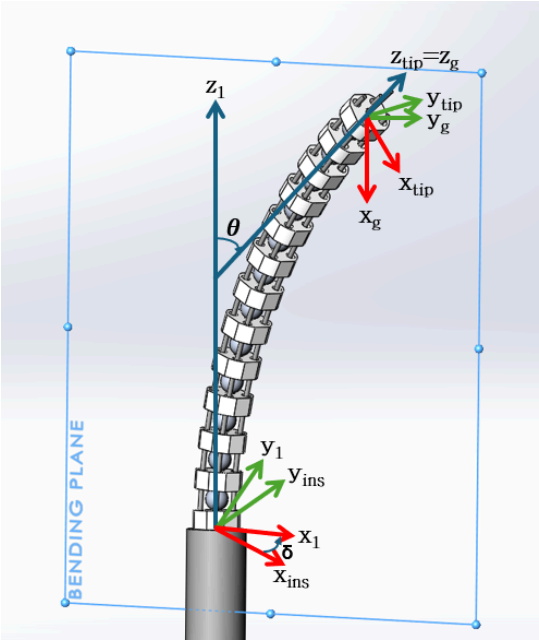


Figure 9. Diagram of the continuum probe bending plane with relevant vector markers

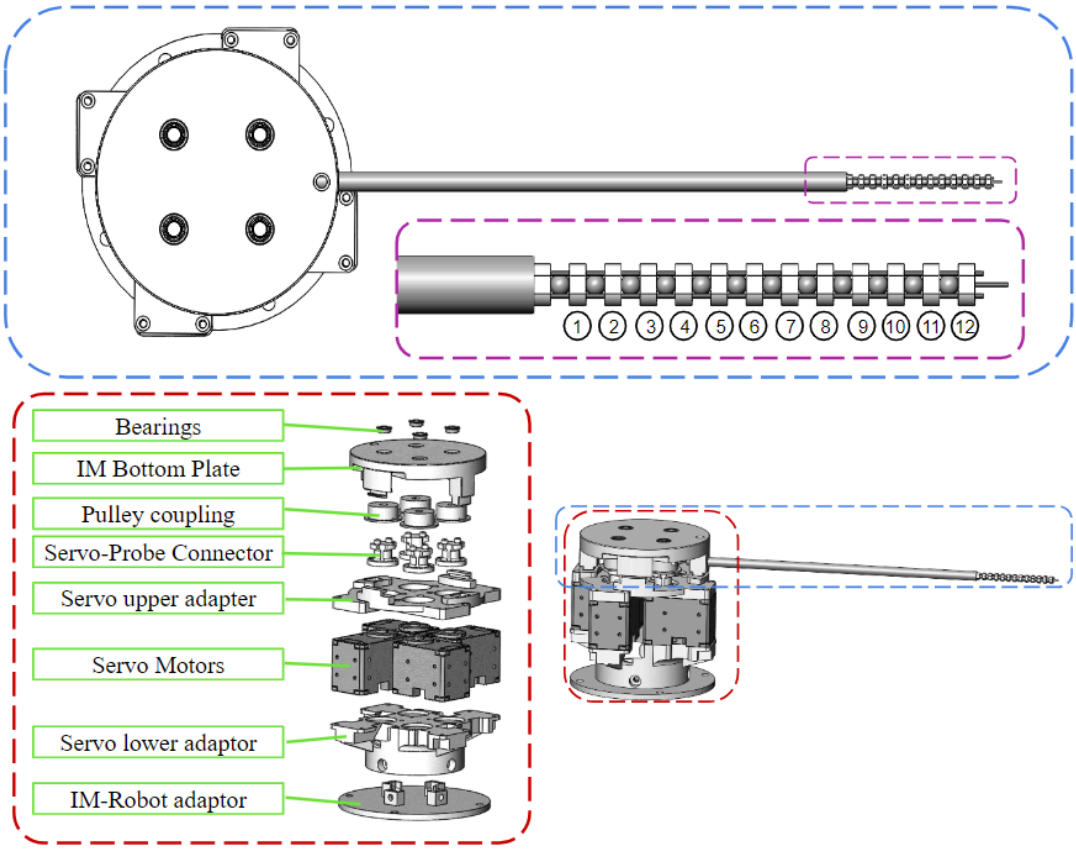


Figure 10. Diagram depicting the Instrument Module viewed from various angles with a detailed exploded view

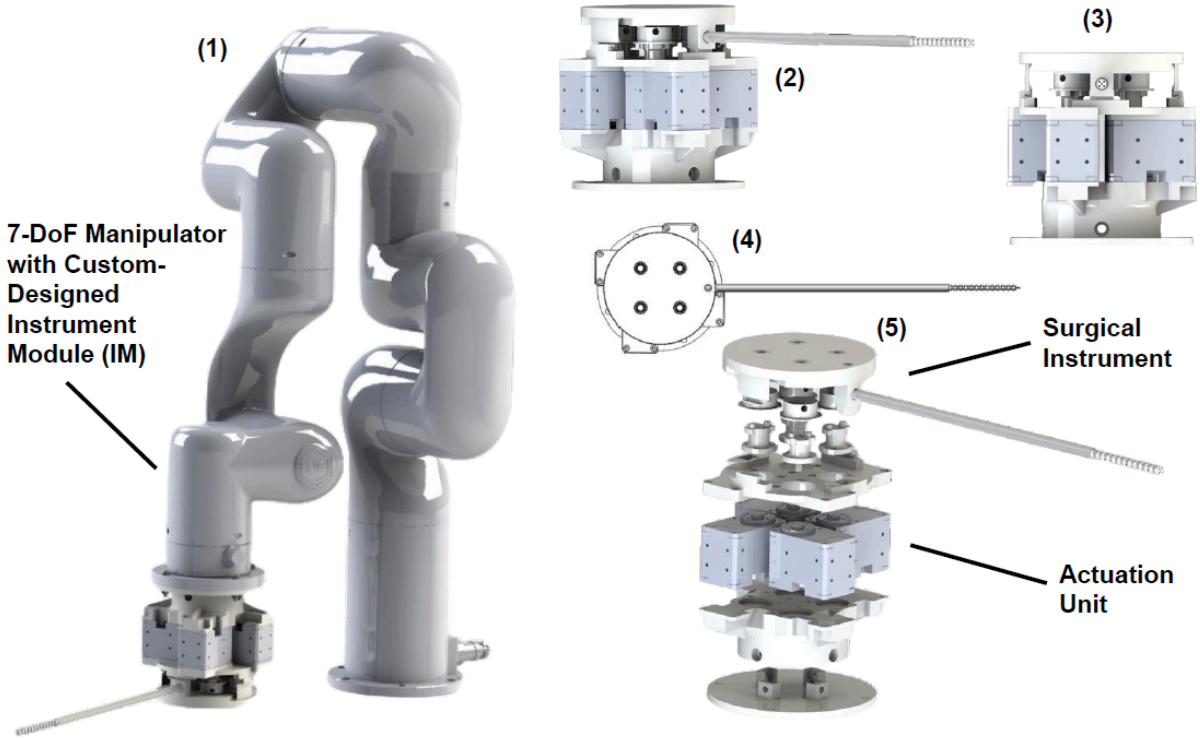


Figure 11. Diagram depicting the rendered Instrument Module attached to the robot arm with exploded view

(I submitted the following abstract to the Stevens Next-Gen Healthcare Innovators Symposium 2024)

Novel Robotic Surgery Mechanism Utilizing A Continuum Instrument

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Introduction: Minimally invasive surgeries (MIS) have become increasingly popular in the past few decades, with a growing trend toward robotic platforms in recent years. MIS is preferred to traditional surgical procedures as it requires fewer incisions and is strongly correlated with improved patient recoveries [1]. Surgical procedures that benefit from these minimally invasive methods include laparoscopy, arthroscopy, gastrointestinal surgery, neurosurgery, fetal surgery, and ear, nose and throat (ENT) surgery [2]. A challenge that can present itself in these constrained procedures stems from achieving the dexterity necessary of the surgical instrument. Several solutions have been proposed, and this research project focuses on the development of a continuum instrument module. A continuum probe provides significant flexibility and allows for single-port surgeries, in which the Remote Center of Motion (RCM) can be constrained to the exterior of the patient's anatomy. This enables precision in the minimally invasive procedure. This continuum probe is formed with twelve small disks that are separated by spherical joints. Each disk is held in place by four titanium wires passing through symmetrically-spaced channels. A fifth titanium wire is passed through the center of the probe, with the tip representing the edge of the actuation space. Each of the four wires can be individually pulled by a spooling mechanism, actuated by a Servo motor, that enables the continuum probe to bend in one of four directions. By simultaneously actuating a second motor, the probe can operate at any angle within a 360 degree range. This research project explores the feasibility of such a continuum probe and aims to confirm its potential in a healthcare and surgical setting.

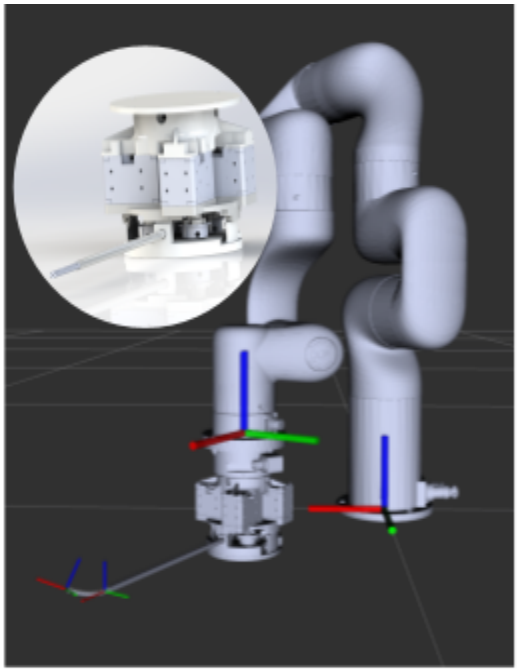


Figure 1: Simulation visualization of a 7-DoF manipulator with continuum instrument in ROS2.

Methods: The continuum instrument went through several major prototype iterations. SolidWorks was utilized to develop an assembly model to achieve the specified goals. This model was 3D printed and constructed to house the Servo motors, wire spooling mechanism, and connect the instrument module to the 7-DOF robot arm. Following the construction of the continuum instrument, a Unified Robotics Description Format (URDF) was created in order to accurately track position of the probe in the simulation space. These simulations are based in the Robot Operating System (ROS), the preferred robotic software platform for this project. Additional testing and validation occurred with Arduino and a TeensyBoard operating system, which allowed for sequential control of each motor and confirmed the mathematics between motor actuation and probe tip movement.

Results & Discussion: Significant progress has been made in the past few months on this surgical continuum robot project. As mentioned, the physical construction is complete, as well as the integration of various software components. Using the actuation controls for the continuum arm, the necessary dexterity has been confirmed. I plan to proceed with the design and test functionality both in simulation and in the surgical kit. The surgical kit simulates a real-world surgical operation, and this will be tested with the 7-DOF robot. This functionality testing is scheduled to be performed in the upcoming weeks, and I believe it will show the success of the instrument and confirm the realistic application of a continuum probe in the surgical field in which significant maneuvrability is required [3]. This novel approach builds on previous knowledge of continuum robotics in the surgical field and single-port surgery procedures.

Acknowledgements: Dr. Long Wang and the Advanced Robot Manipulators Lab, Ehsan Nasiri (PhD)
References: [1] M. Minor and R. Mukherjee, "A Dexterous Manipulator for Minimally Invasive Surgery," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA),* Detroit, MI, 1999, pp. 2057-2064.
[2] K. Xu and N. Simann, "An Investigation of the Intrinsic Force Sensing Capabilities of Continuum Robots," *IEEE Transactions on Robotics,* vol. 24, no. 3, pp. 576-587, June 2008.
[3] A. Degani, H. Choset, A. Wolf, and M. A. Zenati, "Highly Articulated Robotic Probe for Minimally Invasive Surgery," in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), Orlando, FL, 2006, pp. 4167-4172.